

Week 3

Continue with problems. Also note that as you go through the text, Griffiths embeds problems throughout the chapter. Some of the problems have critical results (even though we don't do all the problems), so you should read all the problems and discover important results.

HW Ch. 2A---first assignment from ch. 2

ch. 2 A) problems , 2,6,9, 14 (do inside and outside, give results using vector notation) And a couple of my problems...(which count as 2 points each)

2.2

(a) Find the electric field (magnitude and direction) a distance z above the midpoint between two equal charges, q , a distance d apart (Fig. 2.4). Check that your result is consistent with what you'd expect when $z \gg d$.

(b) Repeat part (a), only this time make the right-hand charge $-q$ instead of $+q$.

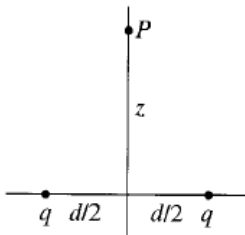


Figure 2.4

2.6 Find the electric field a distance z above the center of a flat circular disk of radius R (Fig. 2.10), which carries a uniform surface charge σ . What does your formula give in the limit $R \rightarrow \infty$? Also check the case $z \gg R$.

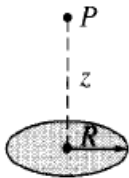


Figure 2.10

2.9

Suppose the electric field in some region is found to be $\mathbf{E} = kr^3\hat{\mathbf{r}}$, in spherical coordinates (k is some constant).

- (a) Find the charge density ρ .
- (b) Find the total charge contained in a sphere of radius R , centered at the origin. (Do it two different ways.)

2.14

Find the electric field inside a sphere which carries a charge density proportional to the distance from the origin, $\rho = kr$, for some constant k . [Hint: This charge density is *not* uniform, and you must *integrate* to get the enclosed charge.]

For problem 2.14 also find the $\mathbf{E}$ field outside the sphere.

Colbert-A. (Label it as "Colbert-A")

A flat circular loop of charge with radius R lies in the xy plane centered on the origin. The linear charge density is equal to $A\cos(\phi)$ where the angle is the usual angle measured with respect to the x axis.

- a) What is the total charge on the loop (show work)?
- b) Set up a general expression to find the $\mathbf{E}$ field vector at any point in space (don't do the integration--set up clearly)
- c) Use your result from part b to find a valid expression for the $\mathbf{E}$ field at any location on the z axis. Be careful.
- d) Find the $\mathbf{E}$ field in the limit as z gets very large compared to R (not infinitely large-- $\mathbf{E}$ does not become zero).

Colbert---B. A cylinder lines up with its central axis along the z axis. The volume charge density is $=k\cos(s/b)$ (s is radial cylindrical coordinate) up to $s=b$ (b is the outer radius), and the density is zero outside that. Use Gauss's to determine the $\mathbf{E}$ field (vector) both inside and outside the cylinder.

DON'T FORGET YOUR ADDITIONAL "YOU MAKE IT UP PROBLEM" cyop